

PROJECT 1A

SECURITISATION RISK ASSESSMENT DASHBOARD

Professional Power BI Dashboard Documentation

Metric	Value
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Project Domain	BFSI / Structured Finance / Credit Risk Analytics
Tool Used	Microsoft Power BI, Power Query, DAX
Submission Type	Professional Project Documentation

This report documents the design, methodology, dashboard pages, key metrics, findings, and recommendations for the Securitisation Risk Assessment Dashboard developed as part of Project 1A.

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1. Executive Summary

The Securitisation Risk Assessment Dashboard was developed to provide a structured risk monitoring framework for an assumed loan-backed securitisation portfolio. The solution uses Microsoft Power BI to consolidate portfolio exposure, delinquency, IFRS 9 Expected Credit Loss, stress testing, geographic risk, vintage performance, and dynamic monthly loss monitoring into one executive reporting environment.

The dashboard allows risk analysts, credit managers, investors, and management stakeholders to quickly identify portfolio health, emerging risk pockets, and segments requiring intervention. It also demonstrates how BI tools can support structured finance and credit-risk decision-making beyond basic reporting.

Metric	Value
Dashboard Pages	7 analytical pages
Total EAD	326.16M
Total ECL	11.66M
Current Balance	317.79M
Net Loss	879K
Main Risk Driver	IFRS 9 Stage 3 / high-PD loans

Executive Interpretation

The portfolio appears broadly stable at an aggregate level, but the dashboard highlights important risk concentrations. Stage 3 loans contribute the majority of expected credit losses, while selected regions and states show higher risk indicators. Stress testing demonstrates that ECL rises materially under adverse scenarios, making ongoing monitoring necessary.

2. Business Context and Problem Statement

Securitisation involves pooling financial assets such as loans and converting them into securities. For banks, NBFCs, and investors, the performance of the underlying loan pool determines the quality of the securitised instrument. Risk can remain hidden if reporting only focuses on aggregate balances.

The business problem addressed in this project is the need for a single analytical view that can answer key credit-risk questions: What is the total exposure? Which regions contribute the highest expected loss? Which accounts are becoming delinquent? How does the portfolio behave under stress? Which vintages show deterioration?

- Limited visibility into expected losses and risk stages can delay action.
- Regional concentration risk may expose the portfolio to local economic weakness.
- Delinquency buckets must be monitored before accounts migrate into severe default.
- Stress testing is needed to estimate the effect of adverse economic conditions.
- Vintage analysis helps assess whether the pool is seasoning as expected.

3. Project Objectives

The primary objective of this project is to design a professional dashboard that converts securitisation portfolio data into risk intelligence. The dashboard should support both executive-level review and analytical drill-down.

Metric	Value
Objective 1	Develop a Power BI dashboard for portfolio monitoring
Objective 2	Calculate and analyse EAD, PD, LGD, and ECL
Objective 3	Track 30+, 60+, and 90+ DPD delinquency buckets
Objective 4	Build stress testing views for baseline, mild, and severe scenarios
Objective 5	Identify geographic concentration risk by state and region
Objective 6	Analyse vintage, pool factor decay, recoveries, and net losses
Objective 7	Monitor monthly default, recovery, and collection trends

4. Dataset and Data Preparation

The dashboard uses securitisation-related datasets covering loan-level exposure, delinquency, IFRS 9 classification, vintage performance, and dynamic monthly performance. The fields support analysis at borrower, state, region, stage, vintage, and month levels.

Key Data Fields

Metric	Value
LoanID / BorrowerID	Unique loan and borrower identifiers
Region / State	Geographic segmentation for concentration analysis
CurrentBalance	Outstanding loan balance
EAD	Exposure at Default
PD_Estimate	Probability of Default
LGD_Estimate	Loss Given Default
ECL_Provision	Expected Credit Loss provision
IFRS9_Stage	Stage 1, Stage 2, or Stage 3 credit classification
DelinquencyDays	Days past due for loan-level monitoring
MonthsOnBook	Vintage seasoning period
Recoveries / Defaults	Performance and recovery indicators

Data Preparation Steps

- Loaded data into Power BI using Power Query.
- Validated numeric fields such as balances, EAD, ECL, PD, and LGD.
- Standardised categorical fields such as region, state, and IFRS 9 stage.
- Created calculated measures for key risk metrics.
- Designed visuals to support executive interpretation and drill-down analysis.

5. Data Model and DAX Measures

The data model was designed to support multiple analytical perspectives. Loan-level data supports exposure, ECL, delinquency, and geographic pages. Vintage and dynamic monthly tables support pool performance and time-series monitoring.

Important Measures

Measure	Business Meaning
Total EAD	Measures total exposure at default across the portfolio.
Total ECL	Aggregates expected credit loss provisions.
Average PD	Shows average probability of default for the selected portfolio segment.
Average LGD	Shows expected loss severity if default occurs.
30+ DPD Loans	Counts accounts with delinquency greater than or equal to 30 days.
60+ DPD Loans	Counts accounts with delinquency greater than or equal to 60 days.
90+ DPD Loans	Counts accounts with severe delinquency greater than or equal to 90 days.
Baseline ECL	Represents current expected loss under normal conditions.
Mild Stress ECL	Applies a stress uplift to estimate expected loss under mild deterioration.
Severe Stress ECL	Applies a higher stress uplift for adverse scenario analysis.
Collection Efficiency	Measures recovery and collection performance over time.

The measures were designed to be interpretable so that each metric has a clear business meaning for credit-risk stakeholders.

6. Dashboard Architecture

The report consists of seven analytical pages. Each page focuses on a specific area of securitisation risk management. The dashboard follows a consistent structure with KPI cards at the top and detailed charts below.

Metric	Value
Page 1	Portfolio Overview
Page 2	DPD and Delinquency Analysis
Page 3	IFRS 9 ECL Analysis
Page 4	Stress Testing and Scenario Analysis
Page 5	Geographic Risk Analysis
Page 6	Vintage and Static Pool Analysis
Page 7	Dynamic Loss and Performance Analysis

This architecture allows management users to begin with an executive summary and then move into deeper analysis across delinquency, ECL, stress impact, geographic concentration, vintage behaviour, and monthly performance.

7. Portfolio Overview Analysis

The Portfolio Overview page provides an executive-level snapshot of the securitisation portfolio. It summarises total exposure, expected losses, outstanding balance, and net loss. The page also visualises exposure by region, net loss by state, and EAD distribution by IFRS 9 stage.

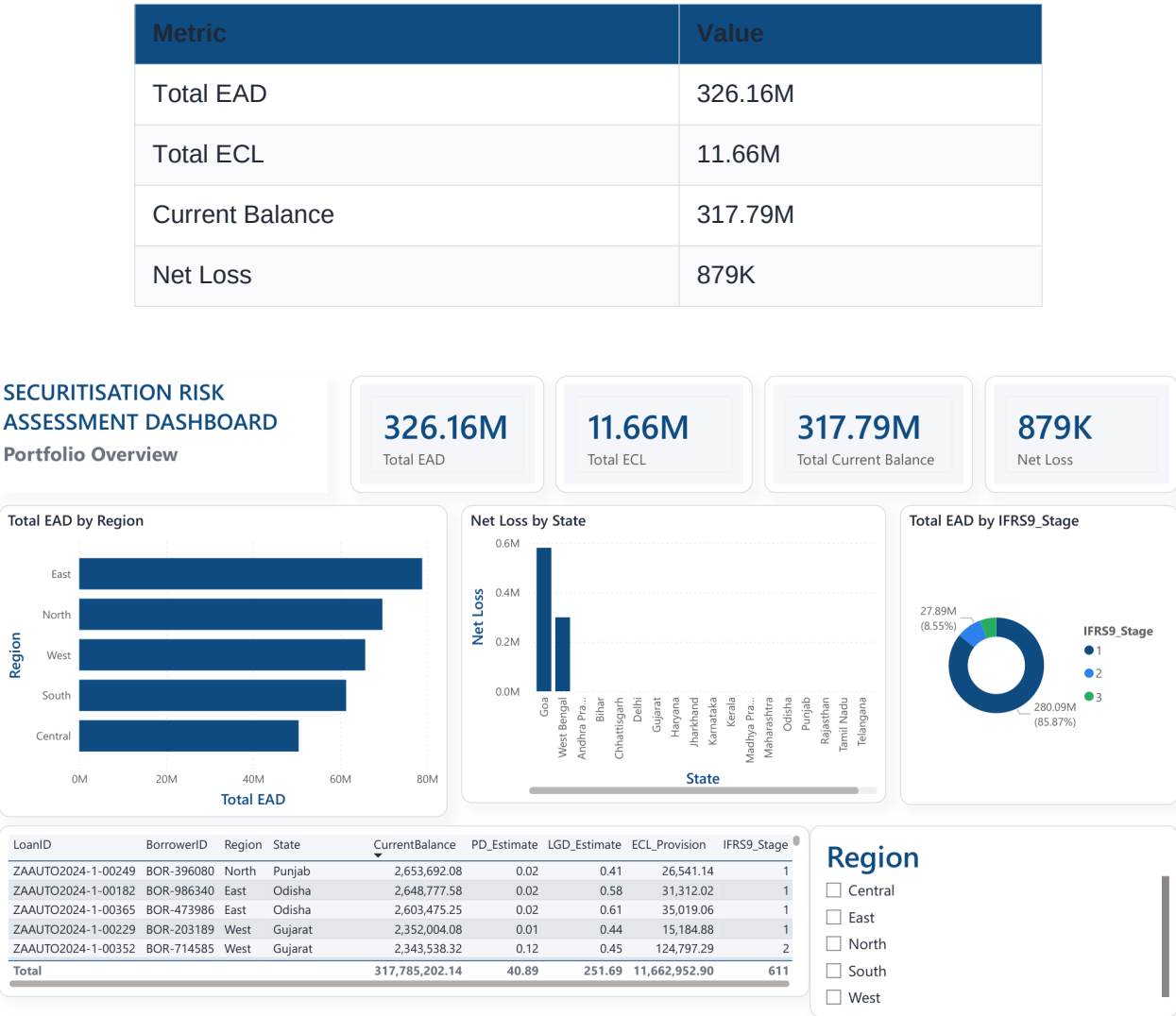


Figure 1: Portfolio Overview Dashboard

Key Insights

- The portfolio has total exposure of 326.16M and expected credit loss of 11.66M.
- Stage 1 loans dominate the portfolio, indicating that most loans are currently performing.
- Regional exposure is highest in the East region, followed by North and West.
- Net loss concentration is visible across selected states, requiring targeted monitoring.

Business Recommendation

Continue tracking high-exposure regions and monitor any migration from Stage 1 into Stage 2 or Stage 3.

8. DPD and Delinquency Analysis

The DPD page focuses on delinquency monitoring. It shows early warning indicators such as 30+ DPD, 60+ DPD, and 90+ DPD loans. The page also compares average delinquency days by region and loan count by DPD bucket.

Metric	Value
30+ DPD Loans	78
60+ DPD Loans	57
90+ DPD Loans	33
Average Delinquency Days	21.21

DPD & DELINQUENCY ANALYSIS

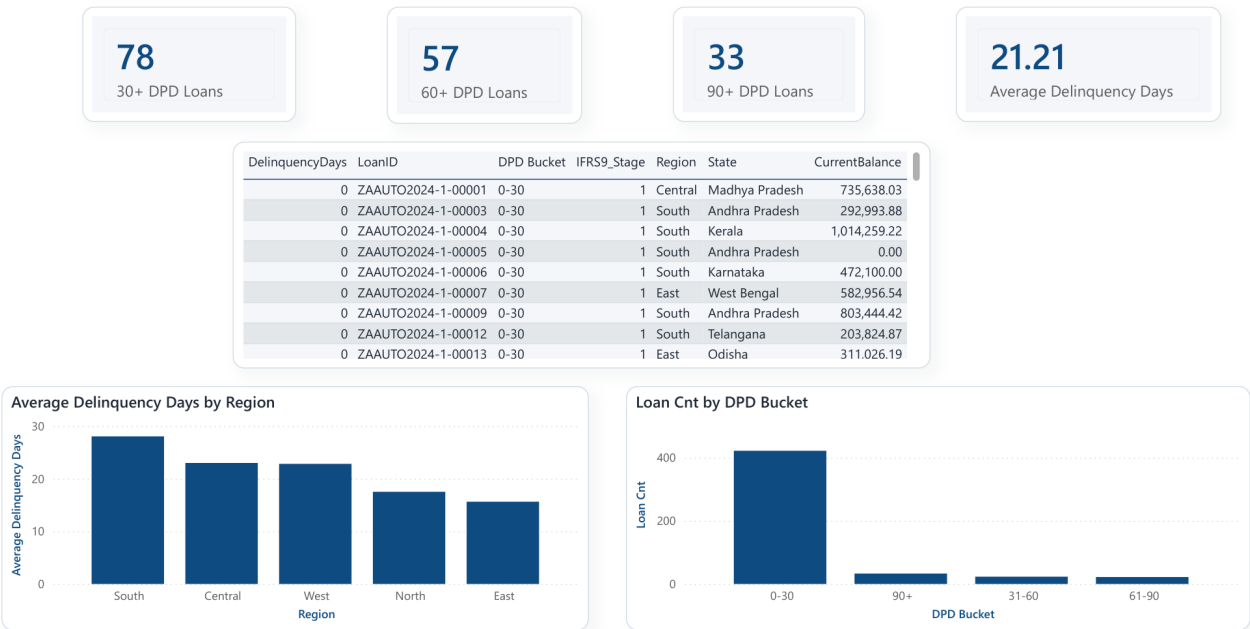


Figure 2: DPD and Delinquency Analysis Dashboard

Key Insights

- The majority of accounts remain in the 0-30 DPD bucket.
- There are 33 severe delinquency accounts in the 90+ DPD category.
- South, Central, and West show higher average delinquency days compared with other regions.
- Delinquency monitoring is essential because DPD migration is an early signal of future losses.

Business Recommendation

Prioritise collections for 60+ and 90+ DPD accounts and create early-warning alerts for accounts moving into 30+ DPD.

9. IFRS 9 ECL Analysis

The IFRS 9 page evaluates credit risk using PD, LGD, EAD, and ECL. The page includes a PD versus LGD risk distribution, ECL distribution by IFRS 9 stage, and ECL by region.



Figure 3: IFRS 9 ECL Analysis Dashboard

Key Insights

- Stage 3 loans contribute the majority of expected credit losses.
- South and North regions exhibit higher ECL contribution compared with other regions.
- The PD versus LGD scatter plot shows clear segmentation between low-risk and high-risk loan groups.
- High-PD and high-LGD accounts should be reviewed more frequently.

Business Recommendation

Strengthen monitoring of Stage 2 and Stage 3 loans and focus on accounts with both high PD and high LGD.

10. Stress Testing and Scenario Analysis

The stress testing page estimates how expected credit losses change under baseline, mild stress, and severe stress scenarios. The page includes scenario comparison charts, incremental ECL impact, and stage-wise ECL exposure.

Metric	Value
Baseline ECL	11.66M
Mild Stress ECL	12.83M
Severe Stress ECL	14.00M
Stress Impact	1.17M

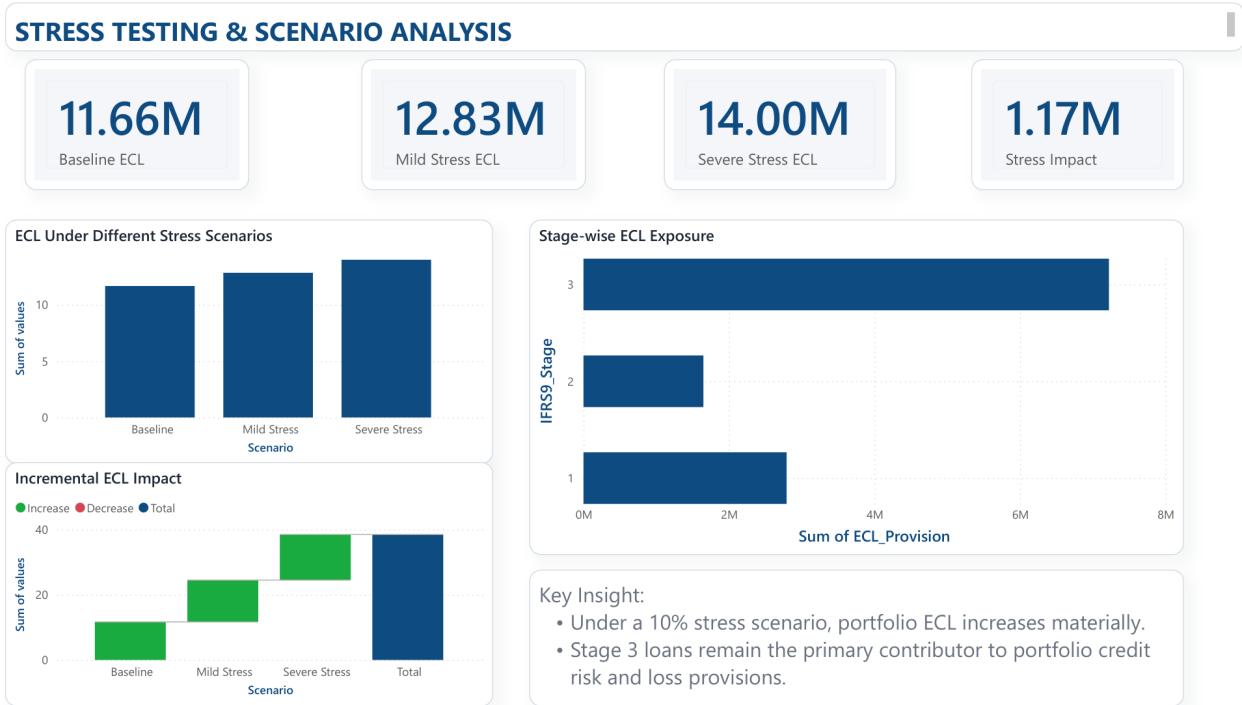


Figure 4: Stress Testing and Scenario Analysis Dashboard

Key Insights

- Expected credit loss increases from 11.66M to 12.83M under mild stress.
- Under severe stress, ECL rises to 14.00M.
- Stage 3 remains the largest contributor to ECL under stressed conditions.
- Stress testing supports capital planning and investor communication.

Business Recommendation

Maintain scenario-based monitoring and evaluate whether stressed ECL levels remain within acceptable risk appetite.

11. Geographic Risk Analysis

The Geographic Risk page evaluates exposure and risk concentration across states and regions. It includes ECL by state, LGD by state, PD by state, and ECL by region.



12. Vintage and Static Pool Analysis

The Vintage and Static Pool page analyses seasoning behaviour through remaining pool trend, pool factor decay, cumulative defaults, recovery trend, net loss evolution, and prepayment behaviour.

Metric	Value
Months on Book	15.36
Marginal Loss	0.03%
Remaining Pool Balance	8.98bn
Total Recoveries	86.86M
Net Loss Rate	0.44%

VINTAGE & STATIC POOL ANALYSIS



Figure 6: Vintage and Static Pool Analysis Dashboard

Key Insights

- The pool factor decay indicates the expected reduction in outstanding balance as the pool seasons.
- Recoveries increase over time before gradually declining as the pool runs off.
- Net loss rate evolution shows how losses accumulate across months on book.
- Vintage analysis is critical for investors evaluating static pool performance.

Business Recommendation

Continue tracking vintage cohorts to identify early deterioration and compare actual performance against expected seasoning curves.

13. Dynamic Loss and Performance Analysis

The Dynamic Loss page tracks monthly operational and credit performance. It includes monthly default rate trend, recoveries versus new defaults, collection efficiency trend, and portfolio balance evolution.

Metric	Value
Monthly Net Loss Rate	0.12%
Collection Efficiency	1.29
Total Monthly Recoveries	4.43M
Monthly Default Rate	0.04



Figure 7: Dynamic Loss and Performance Analysis Dashboard

Key Insights

- Total monthly recoveries amount to 4.43M.
- Collection efficiency of 1.29 indicates active recovery performance.
- Monthly default rate trend helps identify deterioration or improvement in portfolio quality.
- Portfolio balance evolution supports monitoring of balance movement over the reporting period.

Business Recommendation

Monitor monthly spikes in default rates and collection efficiency to identify operational issues early.

14. Consolidated Key Findings

- The portfolio is broadly stable, with total EAD of 326.16M and total ECL of 11.66M.
- Stage 3 loans are the most important driver of expected credit losses.
- DPD analysis shows a manageable but important delinquency population, including 33 loans in the 90+ DPD category.
- Stress testing increases ECL from 11.66M under baseline to 14.00M under severe stress.
- Geographic analysis reveals concentration risk in selected states and regions.
- Vintage analysis confirms the importance of tracking pool factor decay, recoveries, defaults, and prepayments.
- Dynamic monthly analysis supports operational monitoring of collections, defaults, and portfolio balances.

Overall Interpretation

The dashboard demonstrates that a securitisation portfolio can appear healthy at aggregate level while still containing pockets of risk across stages, regions, states, delinquency buckets, and vintages. The report therefore supports both executive summary review and detailed credit-risk investigation.

15. Recommendations

Metric	Value
Stage Monitoring	Review Stage 2 and Stage 3 loans regularly because they carry higher loss risk.
Collections Strategy	Prioritise 60+ and 90+ DPD accounts for collections and recovery action.
Geographic Controls	Track high-PD and high-ECL states through dedicated risk limits and management review.
Stress Testing	Update stress scenarios periodically and compare ECL impact with risk appetite.
Vintage Monitoring	Compare vintage curves with expected pool performance to identify deterioration early.
Predictive Analytics	Introduce ML-based default forecasting in future versions.
Reporting Automation	Publish to Power BI Service with scheduled refresh and stakeholder access controls.

16. Limitations and Future Enhancements

Limitations

- The dashboard is based on assumed or project-level securitisation data rather than a live bank production system.
- Stress scenarios are simplified and use fixed uplift assumptions.
- Predictive modelling is not included in the current version.
- Real-time data refresh and automated alerts are not yet implemented.

Future Enhancements

- Integrate predictive default models using historical behaviour.
- Add automated risk alerts for DPD migration and ECL spikes.
- Deploy dashboard to Power BI Service with role-level security.
- Add investor reporting packs and export-ready summaries.
- Build advanced scenario modelling using macroeconomic variables.

17. Conclusion

The Project 1A Securitisation Risk Assessment Dashboard successfully demonstrates how Power BI can be used to create a professional credit-risk monitoring solution. The dashboard consolidates key risk dimensions such as portfolio exposure, delinquency, IFRS 9 ECL, stress testing, geographic concentration, vintage performance, and dynamic monthly loss monitoring.

The solution provides actionable insights for risk teams, portfolio managers, investors, and executive stakeholders. It also demonstrates practical skills in Power Query, DAX, dashboard design, business interpretation, and structured finance analytics.

Overall, the project is suitable for professional submission as it combines technical implementation with business understanding and analytical storytelling.

References

- IFRS 9 Financial Instruments - Expected Credit Loss Framework.
- Microsoft Power BI Documentation.
- Basel Committee Credit Risk and Securitisation Guidelines.
- Credit Risk Management and Structured Finance Reporting Practices.
- Project 1A Dashboard Export and Power BI Report Assets.